Surprising Findings on Energy Distance in ICA

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Based on Qingyuan's suggestion, I re-ran the experiments about ICA using code that the group is already familiar with—specifically, Qingyuan's `sem2` class and Xinwei's energy loss function. To explore whether energy distance is a feasible loss function for estimating the mixing matrix in ICA, I constructed a simple toy example in the **noise-free case** ($X = AS$). In this setup, I assume **full knowledge of the latent sources**: the number of components, their distributions, and their order. This allows me to directly sample the sources from their known distributions, apply a linear mixing matrix A, and obtain samples from the model. I then compare these samples with the observed data X using energy distance as the loss function.

Importantly, there are **no neural networks involved** in this experiment—the only parameter being estimated is the mixing matrix A, and the loss is a direct function of it. Because ICA is only identifiable up to permutation and scaling of the columns of A, this toy setup—where the latent sources and their order are known—removes those ambiguities.

To evaluate estimation accuracy, I compute the **Amari error** of $\hat{A}$ and compare the result with that of **FastICA**. The formula for the Amari error between the estimated mixing matrix $\hat{A}$ and the ground truth A_{true} is:

$$\text{Amari}(\hat{A}, A_{true}) = \frac{1}{2d}\left(\sum_{i=1}^{d}\left(\frac{\sum_{j=1}^{d}|P_{ij}|}{\max_j |P_{ij}|} - 1\right) + \sum_{j=1}^{d}\left(\frac{\sum_{i=1}^{d}|P_{ij}|}{\max_i |P_{ij}|} - 1\right)\right),$$

where $P = \hat{A}^{-1}A_{true}$. While this comparison may not be completely fair (since FastICA does not assume knowledge of the source distributions), it provides a useful benchmark for evaluating how practical energy distance is as a loss function for estimating the mixing matrix.

I've created a GitHub branch named `molly_test`, where the code and results for this experiment can be found in the file `3D ICA (S known).ipynb`.

The results show that using energy distance as a loss function leads to **very poor estimation of** A—far worse than that of FastICA. However, what's surprising is that the estimated model still produces samples $\hat{X} = \hat{A}S$ whose distribution is **almost indistinguishable** from the true observed data $X = A_{\text{true}}S$ under energy distance. From the training results, it is evident that s_1 and s_2 have become very similar.

Since the data is 3D, it's hard to directly visualize the distributional similarity between the true and generated samples. So I sampled 25,000 points from both the true and generated distributions and computed the empirical energy distance with respect to the training data.

The resulting values show that the two distributions are extremely close, despite $\hat{A}$ being very different from A_{true}.

To further verify the similarity between $\hat{X}$ and X, I plotted **histograms** of each individual dimension. The histograms confirm that for each coordinate, the marginal distributions of $\hat{X}$ and X are nearly identical. I also selected some **2D projections** and plotted scatter plots, which show that in the high-density regions, the sample distributions align very well. However, **the tails of the distributions are not well captured**—$\hat{X}$'s tails deviate from those of X.

During my discussion with Qingyuan this afternoon, we considered that using **standard Euclidean (L2) distance** to calculate distances between all pairs of points in energy distance might be inappropriate. In ICA, we heavily rely on **independence and non-Gaussianity**, so it may be more appropriate to use a different distance metric that reflects these structural properties. For example, we could try designing a custom distance metric that captures **higher-order moments** (e.g., **skewness** or **kurtosis**) and puts more weight on how well the **tails** of the distributions match.